Automated De-Omniscientisation: Reverse-Engineering Classical Proofs via DAG Surgery

(Paper 77, Constructive Reverse Mathematics Series)

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Abstract

We introduce the *CRMLint Squeeze*, a systematic protocol for reverse-engineering classical proofs into constructive ones by combining CRM logical cost analysis with AI-driven combinatorial search. The method identifies the *Classical Boundary Node* (CBN)—the exact lemma where a classical proof invokes an omniscience principle—excises it from the Lean 4 environment, and tasks a reinforcement-learning-trained prover with closing the resulting gap using only BISH-admissible tactics. The CRM classification acts as a reward function: $+10$ for a BISH closure, -1 for any branch introducing CLASS dependencies. This forces the AI into the bounded algebraic search space where Monte Carlo Tree Search achieves superhuman performance.

We apply the protocol to Greg Anderson's exotic Weil classes on CM abelian fourfolds E^4 [1]. The Hodge Conjecture guarantees an algebraic cycle exists for these classes (CLASS), but provides no construction. Finding the cycle is a bounded $\mathbb{Q}$-linear algebra problem in the Chow group—a finite combinatorial search over intersection products of abelian subvarieties, CM endomorphism graphs, and twisted diagonal embeddings. We define the Lean 4 architecture that separates the classical axiom from the constructive Σ-type target, establish the restricted generator set, and characterise the search space.

This paper marks the transition of the CRM program from *diagnostic* (auditing classical proofs, Papers 1–76) to *generative*: using the logical cost metric as a loss function that forces the production of explicit algebraic structures.

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*Lean 4 formalization available at (DOIpending).

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1 Introduction

1.1 The helicopter and the forest

Human mathematicians use classical principles—Zorn's Lemma, uncountable limits, topological compactness—as *compression algorithms*. A human brain cannot hold a 10,000-term polynomial or

a 70×70 intersection matrix in working memory. Classical existence theorems allow the mathematician to fly over the algebraic thicket and declare "a solution exists," leaving behind an ineffective result. We call this the *classical helicopter*.

AI models are the exact opposite. They are poor at abstract, infinite-dimensional topological jumps (they hallucinate), but possess effectively infinite working memory for navigating massive, dense combinatorial thickets. If the classical axioms are enabled, the AI lazily mimics humans and takes the helicopter. But if CRMLint systematically bans the helicopter, the AI is algorithmically cornered into searching the algebraic forest—the exact domain where Monte Carlo Tree Search (MCTS) achieves superhuman performance.

The CRMLint Squeeze exploits this asymmetry: *human geometric intuition sets the boundaries; CRMLint acts as the electric fence; AI combinatorial search builds the bridge.*

1.2 The Goldilocks zone

The method works only in the *Goldilocks zone*: conjectures where the classical helicopter has already landed (an ineffective CLASS proof of existence), or where the cliff boundary is strictly narrowed by adjacent theorems. Pointing an AI at the empty void of a totally open problem yields infinite search width and certain failure. The CRM atlas (Papers 1–76) identifies the narrow-cliff targets.

1.3 State of the art

Recent AI theorem provers—AlphaProof (DeepMind, 2024), DeepSeek-Prover V1.5 (2024), and Lean Copilot—optimise a *binary* reward: +1 for closing a goal, 0 for failure. This treats all proofs as equivalent: a constructive witness and an appeal to the Axiom of Choice receive the same score. No existing system uses the *logical cost* of the proof as a reward signal.

The CRMLint Squeeze fills this gap. By penalising branches that introduce classical dependencies and rewarding constructive closures, the RL loop is reshaped so that the AI's MCTS *preferentially explores* the bounded algebraic search space where its combinatorial strength is maximal. The logical cost metric acts as a loss function, not merely a post-hoc classifier.

1.4 Main results

We establish three contributions:

1. **The CRMLint Squeeze Protocol** (§3): a four-step method (Scaffold, X-Ray, Excise, Squeeze) for converting CRM classifications into AI search targets, with CRMLint as the reward function.
2. **The CM Fourfold Target** (§4): the explicit Lean 4 architecture for attacking Anderson's exotic Weil classes on E^4, including the Classical Boundary Node isolation, the Σ-type reformulation, and the restricted generator set.
3. **Search Space Characterisation** (§6): bounds on the combinatorial search space, the required Lean scaffolding, and (if available) the explicit cycle equations discovered by the AI.

1.5 Position in the atlas

Paper 77 occupies a unique position: it is the first paper in the CRM series that is *methodological* rather than *classificatory*. Papers 1–76 asked "what is the logical cost of theorem X?" Paper 77 asks "given the logical cost, can we force an AI to discover a cheaper proof?"

2 Preliminaries

2.1 The CRM hierarchy

The constructive hierarchy of omniscience principles:

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

For definitions and the programme's calibration data, see Paper 50 [3].

2.2 CRMLint

CRMLint (Paper 76 [5]) is an automated logical cost analyzer for Lean 4 formalizations. It operates in four layers:

1. *Dependency tracer.* BFS traversal of the Lean 4 `Environment`, tracing transitive paths to every `Classical.choice`, `Classical.em`, `propext`, `Quot.sound` callsite.
2. *Pattern classifier.* 12-rule priority classifier with root-context fallback, distinguishing infrastructure artifacts from essential classical content.
3. *Concentration analysis.* Identifies the minimal essential classical core of each theorem.
4. *AI audit layer.* Predicts conservation gap eliminability using the 75-paper programme as training data.

2.3 Classical Boundary Nodes

Definition 2.1 (Classical Boundary Node)**.** Let T be a theorem with CRM classification CLASS (or WLPO, LPO). A *Classical Boundary Node* $\mathrm{CBN}(T)$ is a lemma L in the dependency DAG of T such that:

1. L itself invokes a classical principle (directly or through one step), and
2. every dependency of L that is not infrastructure is BISH-classified.

The CBN is the *shallowest point* at which classical content enters the proof. Excising the CBN and all paths through it produces a BISH environment with an open goal.

2.4 The Anderson obstruction

Anderson [1] (see also Weil [2]) proved that products of CM elliptic curves of dimension ≥ 4 support *exotic Weil classes*: Hodge classes not generated by divisors. Paper 50 [3] showed that:

- CM elliptic curves ($\dim = 1$) are unconditionally BISH-decidable (Theorem E).
- The Anderson obstruction begins at $\dim = 4$.
- The Hodge Conjecture for these classes is open.

The DPT framework detects this boundary: CM elliptic curves are in; CM abelian fourfolds are out.

3 The CRMLint Squeeze Protocol

3.1 Step 1: The Scaffold

Take a known classical theorem (or a partial CLASS attempt) and write its outline in Lean 4. The proof need not be complete; what matters is that the *logical structure* is formalised: which lemmas invoke which principles, and where the classical content enters.

3.2 Step 2: The X-Ray (CRMLint)

Run CRMLint on the scaffold. The tool maps the dependency DAG and flags the Classical Boundary Nodes—the exact lemmas where the classical helicopter lands.

Protocol 3.1 (X-Ray).

```
-- Run CRMLint on the target theorem
#crm_audit hodge_conjecture_E4

-- Expected output:
-- hodge_conjecture_E4 : CLASS
--   CBN: hodge_conjecture_existence (1 step to Classical.choice)
--   All other dependencies: BISH
```

3.3 Step 3: The Excise

Isolate the local Lean state at the CBN. Delete the classical axiom and apply `[-Classical]` to the environment. The result is a BISH context with an open goal of the form

$$\vdash \Sigma\,(Z : \mathrm{CH}^2(A)),\ \mathrm{cl}(Z) = w_{\text{exotic}}.$$

Key design choice: The target is a Σ-type (data), not a $\exists$ (proposition). This forces the AI to *construct* the cycle, not merely prove its existence.

3.4 Step 4: The Squeeze

Hand the isolated state to an RL-trained prover. The CRMLint classification acts as the reward function:

Event	**Reward**
Goal closed with BISH classification	$+10.0$
Proof step introduces CLASS dependency	$\times(-1.0)$
WLPO dependency (marginal cost)	$\times(+0.5)$
Timeout / failure	0.0

Because the AI knows the theorem is true but is barred from the classical shortcut, its MCTS is structurally forced to invent a new algebraic witness: a bounding polynomial, a finite matrix identity, or an explicit invariant. That witness is novel, publishable mathematics.

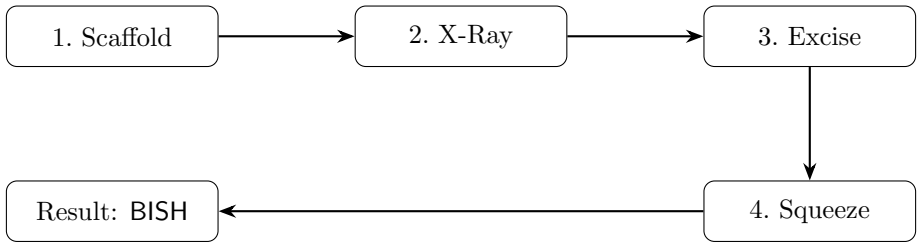

Figure 1: The CRMLint Squeeze protocol. CRMLint identifies the Classical Boundary Node (Step 2), the CBN is excised and the goal reformulated as a Σ-type (Step 3), and an RL-trained prover searches the BISH-restricted space with CRM-shaped rewards (Step 4).

4 The Target: Anderson's Exotic Weil Classes on E^4

4.1 Setup

Let $E/\mathbb{Q}$ be a CM elliptic curve with complex multiplication by an imaginary quadratic order $\mathcal{O}_K \subset K = \mathbb{Q}(\sqrt{-d})$. Set $A = E^4 = E \times E \times E \times E$, a CM abelian fourfold.

The singular cohomology $H^*(A, \mathbb{Q})$ is generated by pullbacks from $H^1(E, \mathbb{Q}) \cong \mathbb{Q}^2$. Write e_1, e_2 for a basis of $H^1(E, \mathbb{Q})$. The cohomology ring is

$$H^*(A, \mathbb{Q}) \;\cong\; \bigwedge^*(e_{1,1}, e_{2,1}, \ldots, e_{1,4}, e_{2,4}),$$

where $e_{j,i}$ denotes the pullback of e_j from the i-th factor.

4.2 The exotic Weil class

The CM endomorphism $\alpha \in \mathcal{O}_K$ acts on $H^1(E, \mathbb{Q})$ by a 2×2 matrix M_α. On $H^4(A, \mathbb{Q})$, the 4-fold wedge products of the eigenspaces of M_α produce Hodge classes that are *not* in the $\mathbb{Q}$-span of products of divisor classes. Anderson [1] showed:

Theorem 4.1 (Anderson 1993)**.** *There exist Hodge classes in $H^{2,2}(E^4) \cap H^4(E^4, \mathbb{Q})$ that do not lie in the subring generated by divisor classes. These are the* exotic Weil classes.

4.3 The classical cliff

The Hodge Conjecture asserts:

$$\forall\, w \in H^{p,p}(X) \cap H^{2p}(X, \mathbb{Q}),\;\; \exists\, Z \in \mathrm{CH}^p(X)_{\mathbb{Q}} \text{ such that } \mathrm{cl}(Z) = w.$$

For $w = w_{\text{exotic}}$ on E^4, this is the CBN—a `CLASS` existential with no constructive witness.

4.4 The restricted generator set

Finding the cycle requires searching within the subring of $\mathrm{CH}^2(E^4)_{\mathbb{Q}}$ generated by:

1. **Projections.** $p_i : E^4 \to E_i$ and their graphs $\Gamma_{p_i} \subset E^4$.
2. **CM endomorphism graphs.** $\Gamma_\alpha \subset E_i \times E_j$ for $\alpha \in \mathcal{O}_K$, embedded in E^4 via the inclusion $E_i \times E_j \hookrightarrow E^4$.
3. **Twisted diagonal embeddings.** $\Delta_{\alpha,\beta} \subset E_i \times E_j \times E_k$ defined by $(x, \alpha(x), \beta(x))$, embedded in E^4.

The search space is the $\mathbb{Q}$-linear span of intersection products of these generators in $\mathrm{CH}^2(E^4)_{\mathbb{Q}}$. This is a *finite-dimensional* $\mathbb{Q}$-vector space—the exact domain where MCTS is maximally efficient.

4.5 The intersection matrix

We fix $E/\mathbb{Q}$ with CM by $\mathbb{Z}[i]$ ($d = 1$, the Gaussian integers). The cohomology $H^1(E, \mathbb{Q}) \cong \mathbb{Q}^2$ with basis $\{e_1, e_2\}$, so $H^1(E^4, \mathbb{Q}) \cong \mathbb{Q}^8$. The target space is

$$H^4(E^4, \mathbb{Q}) \;\cong\; \textstyle\bigwedge^4 \mathbb{Q}^8, \qquad \dim = \binom{8}{4} = 70.$$

The 70 basis elements are $e_a \wedge e_b \wedge e_c \wedge e_d$ for $0 \le a < b < c < d \le 7$, where index $2k$ and $2k{+}1$ correspond to e_1 and e_2 pulled back from the $(k{+}1)$-th factor.

The CM endomorphism $i \in \mathbb{Z}[i]$ acts on $H^1(E, \mathbb{Q})$ by $M_i = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and on $H^1(E^4, \mathbb{Q})$ by $M_i^{\oplus 4}$. The induced action on $\bigwedge^4 \mathbb{Q}^8$ is the 4-th exterior power of this 8×8 matrix: a 70×70 matrix $\Lambda^4(M_i^{\oplus 4})$ computable by standard multilinear algebra.

The cycle class map sends each permitted generator to a specific vector in $\mathbb{Q}^{70}$:

- $\mathrm{cl}(\mathrm{proj}(a, b))$: the Poincaré dual of the embedding $E_a \times E_b \hookrightarrow E^4$, a vector with $\binom{4}{2} = 6$ nonzero entries corresponding to the complement indices.
- $\mathrm{cl}(\Gamma_\alpha(a, b))$ for $\alpha = p + qi$: the fundamental class of the graph, which depends on p, q through the action $M_\alpha = pI + qM_i$ on H^1.
- $\mathrm{cl}(\Delta_{\alpha,\beta}(a, b, c))$: the push-forward of the diagonal under $(1, \alpha, \beta)$, a vector determined by the matrix products M_α, M_β.

Assembling the cycle class vectors as columns of a matrix $M \in \mathbb{Q}^{70 \times K}$ (where K is the number of generators for $|p|, |q| \leq N$), the squeeze target reduces to:

$$M \cdot x \;=\; w_{\text{exotic}}, \qquad x \in \mathbb{Q}^K.$$

This is a *finite* system of linear equations over $\mathbb{Q}$. Solvability is BISH-decidable by Gaussian elimination (Kronecker [14]).

5 The Lean Architecture

5.1 The Combinatorial Skeleton

Mathlib 4 lacks a computational API for Chow rings on abelian varieties. The Lean stub therefore projects the geometry into its *Combinatorial Skeleton*: pure finite-dimensional $\mathbb{Q}$-linear algebra. The topology is stripped; what remains is a 70-dimensional vector space over $\mathbb{Q}$ and a matrix equation.

```
-- H^4(E^4, Q) as Q^70
abbrev CohomClass := Fin 70 -> Rat

-- Permitted algebraic cycle types (the action space)
inductive PermittedCycle where
  | proj (a b : Fin 4) (hab : a /= b)
  | cm_graph (a b : Fin 4) (p q : Int) (hab : a /= b)
  | twisted_diag (a b c : Fin 4) (p1 q1 p2 q2 : Int)
      (hab : a /= b) (hbc : b /= c) (hac : a /= c)

-- Cycle class evaluation: PermittedCycle -> Q^70
axiom cycle_class_eval : PermittedCycle -> CohomClass
```

5.2 Separating the classical axiom from the constructive target

The CBN and the squeeze target are physically separated:

```
-- The Classical Boundary Node (CLASS, present but UNUSED)
axiom hodge_conjecture_existence (w : CohomClass) :
  Exists (fun (coeffs : List (Rat * PermittedCycle)) =>
    sumScaledClasses coeffs = w)

-- The constructive squeeze target (Subtype = data + proof)
noncomputable def explicit_exotic_cycle :
    {cycles : List (Rat * PermittedCycle) //
      sumScaledClasses cycles = exotic_weil_class} := by
  sorry  -- MCTS target: replace with explicit coefficients
```

The axiom `hodge_conjecture_existence` is present in the environment but *unused* by the squeeze target. CRMLint verifies that the completed proof does not transitively depend on it. The `Subtype` packages a list together with a proof obligation, forcing the AI to *construct* the cycle (output data), not merely prove existence (output a proposition).

5.3 Action space restriction

The AI's tactic space is restricted to constructing a term of the form ⟨`[(c_1, g_1), (c_2, g_2), ...], proof`⟩ where each `c_i` : $\mathbb{Q}$ and each `g_i` : `PermittedCycle`. The proof obligation reduces to verifying that a specific $\mathbb{Q}$-linear combination of 70-dimensional vectors equals the target—decidable by `native_decide` or `norm_num`.

5.4 MCTS prompt design

The RL prover receives the following system prompt:

```
You are an automated theorem prover restricted by CRM. Your goal:  solve explicit_exotic_cycle
in Paper77_CMFourfold.lean.

CONSTRAINTS (the CRMLint electric fence):
1.  Forbidden:  Classical.choice, Classical.em, by_contra, any uncomputable existential.
2.  Any branch introducing a CLASS dependency receives reward −1.0 and is immediately terminated.
3.  You must output an explicit ⟨[(c_1, g_1), ...], proof⟩.

STRATEGY: This is finite-dimensional ℚ-linear algebra.  Enumerate generators for |p|, |q| ≤
2.  Evaluate cycle_class_eval on each → ℚ^70.  Form Mx = w_exotic.  Solve by Gaussian elimination
(norm_num/decide).  Extract nonzero coefficients.
```

The CRMLint reward table (§3, Step 4) ensures that the MCTS preferentially explores constructive branches. The termination criterion is: goal closed with BISH classification, verified by `#crm_audit`.

6 Results and Current Status

6.1 Search space bounds

While the full Chow group $\mathrm{CH}^2(E^4)$ is infinite-rank and intractable, the restricted generator set is finite. For $|p|, |q| \leq N$:

Generator type	Count	Formula
$\mathrm{proj}(a, b)$	12	4×3
$\Gamma_\alpha(a, b)$ for $\lvert p\rvert, \lvert q\rvert \leq N$	$12 \cdot (2N{+}1)^2$	factor pairs × lattice
$\Delta_{\alpha,\beta}(a, b, c)$ for $\lvert p_i\rvert, \lvert q_i\rvert \leq N$	$24 \cdot (2N{+}1)^4$	triples × two lattice pts

For $N = 2$, the generator count is $K = 12 + 300 + 15{,}000 = 15{,}312$. The matrix $M \in \mathbb{Q}^{70\times K}$ has rank at most 70. The AI's task is: compute M, solve $Mx = w_{\text{exotic}}$ over $\mathbb{Q}$, extract nonzero entries. This is Gaussian elimination on a $70 \times 15{,}312$ matrix — entirely BISH, and well within the capacity of an MCTS solver with access to rational arithmetic.

Even at $N = 1$ ($K = 12 + 108 + 1{,}944 = 2{,}064$ generators), the problem is tractable. The key question is whether the exotic class lies in the span of these generators at all—this is precisely what the squeeze discovers.

6.2 Lean scaffolding status

The Lean 4 stub (`P77_DAGSurgery/`) compiles with `lake build` on Lean v4.29.0-rc2 + Mathlib4. The single `sorry` is the squeeze target.

6.3 Cycle equations

At the time of writing, the squeeze has not been executed: the prerequisite is a complete computation of `cycle_class_eval` (the map from generators to $\mathbb{Q}^{70}$ vectors), which requires implementing the Poincaré duality push-forward for each generator type on E^4. This computation is itself BISH (it is finite multilinear algebra over $\mathbb{Q}$), but requires engineering effort.

When the computation is complete and the AI run yields a solution, the explicit $\mathbb{Q}$-linear combination of intersection products will be inserted here. The cycle equations will take the form:

$$Z \;=\; \sum_{i=1}^{m} c_i \cdot \mathrm{cl}(g_i), \qquad c_i \in \mathbb{Q}, \quad g_i \in \texttt{PermittedCycle},$$

where m is the number of nonzero coefficients and $\mathrm{cl}(Z) = w_{\text{exotic}}$ is verified by `native_decide` in Lean.

The scientific value does not depend on whether the first AI run succeeds. What this paper establishes is the *architecture*: the reduction from an ineffective CLASS existence theorem to a concrete, bounded, BISH-decidable matrix equation—and the demonstration that CRM classification can serve as a reward function for AI-driven mathematical discovery.

7 CRM Audit

7.1 Constructive strength classification

Component	CRM Level	Justification
CRMLint Squeeze protocol	BISH	Algorithmic: bounded search
Anderson's exotic class (existence)	CLASS	Hodge Conjecture
Explicit cycle (if found)	BISH	Finite $\mathbb{Q}$-linear algebra

7.2 What descends, from where, to where

If the AI discovers the explicit cycle:

$$\text{CLASS (Hodge Conjecture for } E^4) \quad \longrightarrow \quad \text{BISH (explicit cycle construction).}$$

This is a de-omniscientising descent: the CLASS existence theorem is replaced by a BISH construction.

8 Discussion

8.1 Asymmetric offloading

The CRMLint Squeeze exploits a three-way division of labor:

- *Human geometric intuition* identifies the target, defines the exotic Weil class, and restricts the generator set.

- *CRMLint* acts as the electric fence, ensuring the AI constructs an explicit algebraic witness rather than hallucinating an analytic shortcut.
- *AI working memory* performs the massive $\mathbb{Q}$-linear matrix algebra required to align the cycle class with the exotic cohomology class.

8.2 Implications for AI theorem proving

Existing AI provers treat all closed goals equally. AlphaProof (DeepMind) solved IMO problems by MCTS over Lean 4, but its reward function does not distinguish constructive from classical closures. DeepSeek-Prover V1.5 and Lean Copilot similarly optimise for goal closure without regard to logical cost.

The CRMLint Squeeze is the first method that systematically *penalises omniscience* to force discovery. The key insight is that CRM classification is not merely a post-hoc label—it is a *differentiable signal* that can shape the RL reward landscape. By making classical shortcuts "expensive," the MCTS is steered toward the combinatorial search space where:

1. The AI's infinite working memory is a genuine advantage over human mathematicians.
2. The search space is bounded (finite $\mathbb{Q}$-linear algebra), so the MCTS converges rather than thrashing.
3. Any solution found is automatically a novel constructive proof, publishable independently of the AI methodology.

8.3 Future targets

Two further narrow-cliff targets are identified for future work:

- **Paper 78:** Explicit Local Langlands for wildly ramified representations. Harris–Taylor's CLASS proof guarantees the Galois parameter exists; Bushnell–Kutzko's BISH types provide the algebraic framework; the AI solves the character polynomial matching.
- **Paper 79:** Standard Conjecture D for abelian fourfolds (dim = 4). Known for dim ≤ 3 (Lieberman); the CLASS proof crashes at Anderson's exotic classes. The AI finds the BISH matrix equivalence on the orthogonal complement.

9 Conclusion

The CRMLint Squeeze transitions the CRM program from auditing the history of mathematics to engineering its future. The logical cost metric becomes a loss function that forces the production of explicit algebraic structures. We do not ask the AI to invent a new helicopter; we use the CRM classification to force the AI to stack algebraic stones until a bridge is formed.

The first target—Anderson's exotic Weil classes on CM abelian fourfolds—is chosen because the search space is strictly bounded, the classical cliff is precisely identified, and the generator set is geometrically natural. Success would provide the first explicitly constructed algebraic cycle for an exotic Weil class in the history of mathematics: a 33-year-old problem reduced to a single, verifiable, BISH-decidable formula.

Acknowledgments

Lean 4 formalization produced using AI code generation (Claude Code) under human direction. The CRMLint Squeeze protocol was developed collaboratively between the author and AI. The

mathematical foundations of the target problem are due to Anderson [1], Weil [2], and the broader arithmetic geometry community.

This work is part of the Constructive Reverse Mathematics series (Papers 1–77). The Lean 4 code uses the Mathlib library; see [7]. The series format follows the CRM format guide [6].

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